

Marking Scheme

2026 · Paper 1 · 105 marks

General Rules

- M / A / PROOF mark types
- **ft.** : 承上題錯誤但方法正確者，可獲後續步驟分。
- Deduction cap : $A(1)=1 \cdot A(2)=1 \cdot B=0$ (u, pp)
- u = 單位遺漏或錯誤；pp = 準確度不符要求。每部分最多扣 1 分。
- 答案正確且步驟合理者可獲滿分。
- 符號書寫略有出入但意思清楚者不扣分。 · Benefit of doubt
- 分數、小數或根式形式均接受，除非題目另有指明。

Section A(1) (35 marks)

1. Change of Subject [B-]

a (3 marks)

M1 $4h + 12k = 9h + 7$

M1 $12k - 7 = 5h$ **ft.**

A1 $h = \frac{12k - 7}{5}$ **ft.**

2. Algebraic Fractions [B]

a (3 marks)

M1
$$= \frac{4(4x - 5) - 3(3x - 2)}{(3x - 2)(4x - 5)}$$
M1 for common denominator

M1
$$= \frac{16x - 20 - 9x + 6}{(3x - 2)(4x - 5)}$$
 ft.

A1
$$= \frac{7x - 14}{(3x - 2)(4x - 5)}$$
 ft.
Accept $\frac{7(x - 2)}{(3x - 2)(4x - 5)}$

3. Quadratic Equations (Applications) [B]

a (3 marks)

M1 $8^2 + (9 + r)^2 = (23 - r)^2$
M1 for Pythagoras' theorem

M1 $64 + 81 + 18r = 529 - 46r$ **ft.**

A1 $r = 6$ **ft.**
u for cm not required

4. Factorization [B]

a (1 mark)

A1 $(3p - 2)(3p + 2)$

b (1 mark)

A1 $q(3p - 2)(p + 1)$

c (2 marks)

M1 $= (3p - 2)(3p + 2) - q(3p - 2)(p + 1)$ **ft.**
ft from (a), (b)

A1 $= (3p - 2)(3p + 2 - pq - q)$ **ft.**

5. Percentages [B]

a (2 marks)

M1 $\text{marked price} \times (1 - 20\%) = 720$

A1 $\backslash 900$
u-1 if \$ sign / unit missing

b (2 marks)

M1 $\text{cost} \times (1 + 20\%) = 720$

A1 $\backslash 600$

6. Inequalities [B+]

a (2 marks)

M1 $6x + 37 \leq 10x + 30$

A1 $x \geq \frac{7}{4}$ ft.

b (2 marks)

M1 $4x - 8 < 21 - x \Rightarrow x < \frac{29}{5}$

A1 $4(\text{i. e. } x = 2, 3, 4, 5)$ ft.

7. Ratios (Applications) [B+]

a (4 marks)

M1 *Let 7x and 3x be the original numbers of adults and children*

M1 $\frac{7x + 13}{3x + 12} = \frac{11}{6}$

M1 $6(7x + 13) = 11(3x + 12)$ ft.

A1 $42(\text{adults})$ ft.
Answer must be 7x = 42, not x = 6

8. Pie Charts [B+]

a (1 mark)

A1 **3**
From the 108° sector (12 girls)

b (2 marks)

M1 $\frac{1(8) + 2(10) + 3(12) + 4(6) + 5(4)}{40}$

A1 **2.7** ft.

c (2 marks)

M1 $\frac{6 + 4}{40}$

A1 $\frac{1}{4}$ ft.

9. Volume and Surface Area of Solids [B+]

a (3 marks)

M1 *Let r be the small radius : $\frac{4}{3}\pi((3r)^3 + r^3) = 1008\pi$*

M1 $\frac{4}{3}(28r^3) = 1008 \Rightarrow r^3 = 27, r = 3$ ft.

A1 $972\pi \text{ cm}^3$ ft.
u-1 if unit missing

b (2 marks)

M1 $4\pi(9^2 + 3^2)$ ft.

A1 $360\pi \text{ cm}^2$ ft.

Section A(2) (35 marks)

10. Partial Variation [A-]

a (3 marks)

M1 *Let $h(x) = c + dx$*

M1 $c - d = -42, c + 4d = 48$ ft.

A1 $h(x) = 18x - 24$ ft.

b (2 marks)

M1 $3x^2 = 18x - 24 \Rightarrow x^2 - 6x + 8 = 0$ ft.

A1 $x = 2 \text{ or } x = 4$ ft.

11. Remainder Theorem [A-]

a (3 marks)

M1 *Let the quotient be $ax + b$: $p(x) = (2x^2 + 3x + 7)(ax + b)$*

M1 $p(1) = 12(a + b) = 60$, $p(-3) = 16(-3a + b) = -48$ ft.

A1 *quotient* $= 2x + 3$ ft.

b (3 marks)

M1 *Discriminant of $2x^2 + 3x + 7 = 0$ is $3^2 - 4(2)(7) = -47 < 0$*

M1 *So $2x^2 + 3x + 7 = 0$ has no real root; only $2x + 3 = 0$ gives a rational root* ft.

A1 *1 rational root* ($x = -\frac{3}{2}$) ft.

12. Stem-and-Leaf Diagrams [A]

a (2 marks)

M1 $n = 20$: $Q_1 = \text{mean of 5th, 6th} = 60 + c$, $Q_3 = 72$

A1 $72 - (60 + c) = 8 \Rightarrow c = 4$ ft.

b (6 marks)

M1 *Mean* : $\frac{(50 + a) + 586 + 657 + (80 + b)}{20} = 69$

M1 $\Rightarrow a + b = 7$ ft.

M1 *Range* $= (80 + b) - (50 + a) = 30 + b - a > 34 \Rightarrow b - a > 4$

A1 $(a, b) = (0, 7)$ or $(1, 6)$ ft.

M1 *(ii) Use the less - spread set $(a, b) = (1, 6)$ to compute s.d.* ft.

A1 *Least possible standard deviation* ≈ 7.34 s ft.
r.t. 7.34

13. Properties of Circles [A]

a (3 marks)

M1 $\angle AED = 110^\circ$; $\widehat{CD} = 2(110^\circ) - 180^\circ = 40^\circ$

M1 $\angle CBF = \angle CBD = \frac{1}{2}\widehat{CD}$ ft.

A1 $\angle CBF = 20^\circ$ ft.

b (5 marks)

M1 $\angle DOC = 40^\circ$; $BC \parallel OD \Rightarrow \angle OCB = 40^\circ$

M1 $\triangle OBC$ isosceles : $\angle BOC = 180^\circ - 2(40^\circ) = 100^\circ$ ft.

M1 *Arc* $= 18 \times \frac{100\pi}{180} = 10\pi$ ft.

M1 *Perimeter* $= 10\pi + 2(18) = 10\pi + 36 \approx 67.4$ cm ft.

A1 $67.4 > 65$, so the perimeter is not less than 65cm (disagree) ft.

14. Congruent and Similar Triangles [A]

a (4 marks)

PROOF (i) $BG \parallel EC$ and $CG \parallel DB$ so $BFCG$ is a parallelogram; $BC = CB$ (common), $\angle GBC = \angle FCB$ and $\angle GCB = \angle FBC$ (alt. $\angle$ s) $\Rightarrow \triangle BCG \cong$

A2 (i) correctly reasoned congruence
2A: deduct 1 for each missing key reason

PROOF (ii) $\angle BFC = \angle DFE$ (vert. opp.); $\angle FBC = \angle FDE$ (alt. $\angle$ s, $BC \parallel AD$) $\Rightarrow \triangle BCF \sim \triangle DEF$ (AAA).

A2 (ii) correctly reasoned similarity

b (4 marks)

M1 $\angle BCF = \angle BGC = \angle BFC \Rightarrow BF = BC = \ell$

M1 $BD = \sqrt{2}\ell$, $DF = BD - BF = \sqrt{2}\ell - \ell$ ft.

A1 $DF = (\sqrt{2} - 1)\ell$ ft.

A1 *From similarity* $AE = (2 - \sqrt{2})\ell \approx 0.586\ell > DF \approx 0.414\ell$; agree ft.

Section B (35 marks)

15. Permutations and Combinations [A]

a (3 marks)

M1 $C_5^{32} - C_5^{11}$ (all minus all - girls)

M1 $= 201376 - 462$ ft.

A1 **200914** ft.

16. Arithmetic Sequences [A]

a (2 marks)

A1 $\alpha = 10^3 = 1000$

A1 $\log \beta = 2(3) - 1 = 5 \Rightarrow \beta = 100000$ ft.

b (4 marks)

M1 *first term 3, common difference 2* ft.

M1 $S_n = \frac{n}{2}[6 + 2(n-1)] = n(n+2)$ ft.

M1 $n(n+2) > 888 \Rightarrow n^2 + 2n - 888 > 0$ ft.

A1 $n > 28.8 \dots \Rightarrow \text{least } n = 29$ ft.

17. Locus [A]

a (2 marks)

PROOF *Join the in-centre to the three vertices; $a = \frac{1}{2}r(CD) + \frac{1}{2}r(DE) + \frac{1}{2}r(EC) = \frac{1}{2}r p$, hence $pr = 2a$.*

A2 *correct proof*

b (5 marks)

A1 (i) *It is the angle bisector of $\angle OHK$ (through H)*

M1 (ii) $OH: 4x - 3y = 0; HK: 8x + 15y - 168 = 0$

M1 $\frac{|4x - 3y|}{5} = \frac{|8x + 15y - 168|}{17}$ ft.

M1 $17(4x - 3y) = 5(8x + 15y - 168)$ (internal bisector) ft.

A1 $9x + 2y - 70 = 0$ ft.

18. Trigonometry in 3-D [A+]

a (5 marks)

M1 *Sine rule in $\triangle ABD$: $\frac{\sin \angle BAD}{BD} = \frac{\sin \angle ABD}{AD}$*

A1 $\angle BAD \approx 51.8^\circ$ ft.
r.t. 51.8

M1 $AB = \frac{AD \sin \angle ADB}{\sin \angle ABD}, AP = AB \cos \angle BAD$ ft.

M1 $CP^2 = AC^2 + AP^2 - 2 AC AP \cos 60^\circ$ ft.

A1 $CP \approx 13.1 \text{ cm}$ ft.
r.t. 13.1

b (2 marks)

M1 $\angle BPC$ requires $BP \perp AD$ and $CP \perp AD$; $AP \approx 9.15 \neq 7.5$ so $CP \nperp AD$

A1 *The claim is incorrect* ft.

19. Quadratic Functions and Circles [A+]

a (1 mark)

A1 $f(3) = 9 + 6k - 18 + 41 - 6k = 32$, so it passes through F

b (11 marks)

M1 (i) $g(x) = f(-x) + 4 = x^2 + (6 - 2k)x + (45 - 6k)$

M1 *Completing the square: $g(x) = (x + (3 - k))^2 + (36 - k^2)$* ft.

A1 (i) $U = (k - 3, 36 - k^2)$ ft.

M1 (ii) *least circle $\Leftrightarrow FO$ diameter $\Leftrightarrow \angle FUO = 90^\circ$*

M1 $\vec{UF} \cdot \vec{UO} = 0$ ft.

M1 *Simplify to $k^4 - 39k^2 - 9k + 162 = 0$* ft.

A1 (ii) $k = 6$ (the only positive root) ft.

M1 (iii) *fixed point of $g: G = (-3, 36)$*

M1 $V = U|_{k=6} = (3, 0)$ ft.

M1 *Circle through $F, O, V: x^2 + y^2 - 3x - 32y = 0$* ft.

A1 *Substituting $G(-3, 36)$ gives $162 \neq 0$; so F, G, O, V are not concyclic* ft.